TECHNICAL SPECIFICATION & ARCHITECTURE MANUAL • V3.2

GeneWise — Technical Document

Full-Stack Next.js 14 & FastAPI Architecture, Bi-LSTM Mathematics, Message-Only Binary PDF Serializer, Section 10 IoT Protocol & Zero-Trust Security

FRONTEND RUNTIME
Next.js 14.2 • React 18.3 • TS 5.3

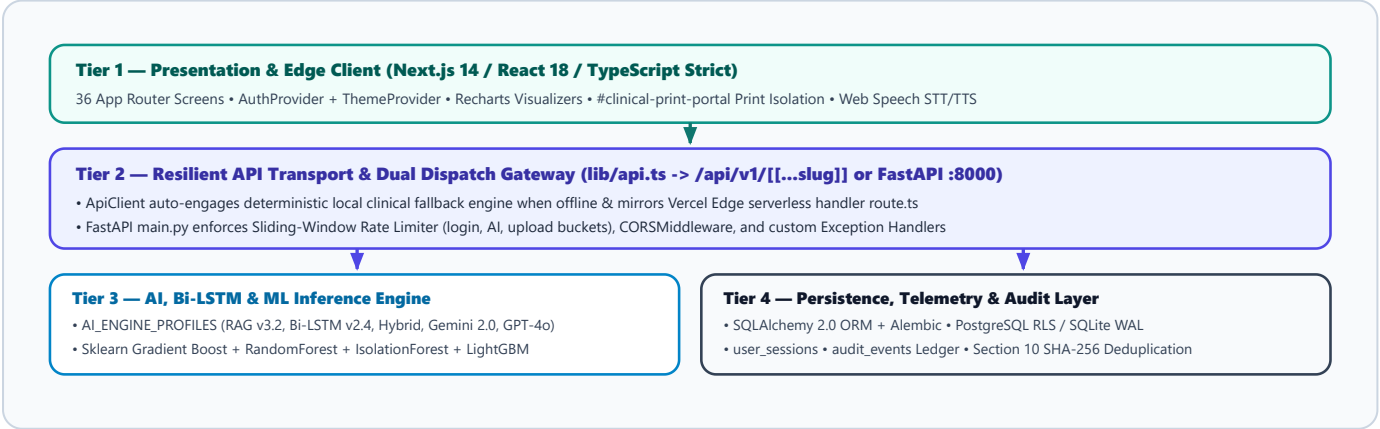
BACKEND & ML STACK
FastAPI • SQLAlchemy 2.0 • Bi-LSTM / Sklearn

SECURITY & VERIFICATION
JWT HS256 • RLS • 37/37 Vitest • 43 Pages

1. Monorepo Structure & Technology Stack

Layer / Directory	Key Packages & Symbols	Engineering Responsibility
`apps/web/app/`	Next.js 14 App Router (36 pages + `/api/v1/[[...slug]]`), Tailwind 3.4, Radix UI, Recharts 2.12, Framer Motion 11	SSR/Static prerendered UI (43 build outputs), Clinical Calm tokens, public documentary pages, interactive pedigree & ML/Bi-LSTM studios.
`apps/web/lib/`	`api.ts` (`ApiClient`), `clinical-lstm.ts`, `export-share.ts`, `auth.tsx`, `genomics-data.ts`	Resilient API client with deterministic clinical fallback engine, client/server Bi-LSTM forward pass, binary PDF 1.4 generator, social share builder.
`backend/app/`	FastAPI >=0.110, SQLAlchemy >=2.0.28, Alembic >=1.13, Passlib/Bcrypt, PyJWT, HTTPx	21 versioned REST routers under `/api/v1`, sliding-window IP rate limiter, IDOR ownership checks, WAL SQLite / PostgreSQL pgvector ORM.
`ml/` & `hardware/`	Scikit-Learn >=1.4.1, NumPy, Pandas, Joblib, Python 3.11 sensor simulators	Tabular Celiac Gradient Booster training/calibration, feature engineering pipelines, and ESP32/Wearable/RPi5 Section 10 telemetry emitters.
`apps/web/__tests__/` & `backend/tests/`	Vitest 5.1, Testing Library React 16.3, JSDOM 29, Pytest 8.1	9 Vitest UI/integration/unit suites (37 tests) + 6 FastAPI Pytest security, workflow, agent, and voice verification modules.

2. 4-Tier Hybrid Serverless & Microservices Topology



3. Bi-LSTM Longitudinal Sequence Engine & AI Routing

The sequential inference engine in `apps/web/lib/clinical-lstm.ts` executes a deterministic **2-Layer Bidirectional LSTM** pass across 5 historical clinical visits (`V-4 (-12m)` through `V0 (Now)`) for 6 supported presets (`hba1c`, `glucose`, `ttg_iga`, `apob`, `ferritin`, `systolic_bp`).

Mathematical Cell & Fusion Equations

```
x_norm[t] = (x[t] - mean) / (scale * 0.25 + 1e-6)
f_t = sigmoid(0.75 * h_prev - 0.35 * x_t + 0.45)
i_t = sigmoid(0.95 * x_t + 0.30 * h_prev + r_t)
c~_t = tanh(0.85 * x_t + 0.42 * h_prev)
c_t = f_t * c_prev + i_t * c~_t
o_t = sigmoid(0.68 * x_t + 0.52 * c_t + 0.35)
h_t = o_t * tanh(c_t)
biFusion = tanh(0.65 * h_fwd - 0.25 * h_bwd)
V(+90d)  = V0 + weightedDelta * (0.82 + |biFusion|
* 0.16)
```

Registered Production AI & ML Models

Model ID	Algorithm	Benchmark
`celiac_risk_v1`	GradientBoosting	**0.962** AUC
`bilstm_seq_v2`	2-Layer Bi-LSTM	**0.958** AUC
`cardio_risk_v2`	RandomForest	**0.941** AUC
`anomaly_iso_v1`	IsolationForest	**0.945** AUC
`prs_genomic_v3`	LightGBM + SHAP	**0.974** AUC

AI Assistant Provider Matrix (`AI_ENGINE_PROFILES`)

Provider ID	Engine Name & Model Tag	Architecture & Latency	Primary Strengths
`clinical_evidence`	GeneWise Supervisor RAG v3.2 (`pubmed-grounded-v3.2`)	Multi-Agent PubMed RAG + Guideline Verifier (~190 ms)	Default offline/online PubMed PMID grounding, 96.4% citation faithfulness.
`lstm_temporal`	Bi-LSTM Temporal Biomarker Engine (`bilstm-biomarker-v2.4`)	2-Layer Bi-LSTM (128u) + Attention (~42 ms)	Computes longitudinal biomarker velocity, 30d/90d horizon & gate weights.
`hybrid_rag_lstm`	Hybrid RAG + Bi-LSTM Fusion v3 (`hybrid-fusion-v3.0`)	Bi-LSTM Trajectory + PubMed RAG (~215 ms)	Fuses sequential trend predictions with ADA/ACG/AHA intervention thresholds (0.967 AUC).
`gemini` / `openai`	Gemini 2.0 Clinical / GPT-4o Reasoner	External LLM Multimodal / High-Context (~310–340 ms)	Complex differential framing & polypharmacy reasoning when API keys configured.

Categorized Inquiry & Prompt Architecture

Implemented in `apps/web/lib/clinical-chat-prompts.ts` across 8 specialty filters (*Celiac & Gastroenterology, Endocrinology, Cardiometabolic, Medical Genetics, Bi-LSTM Sequence, Preventive Longevity, Hepatology, Biomarker Triaging*).

- 12 evidence-tagged inquiry starters + patient priority shortcuts (e.g., *Rajesh Kumar CMP-14*).
- Per-reply dynamic `follow_up_questions[]` chips tailored to detected biomarker domain.

Evidence Grounding & Safety Escalation

- Queries matching high-acuity symptoms (chest pain, acute dyspnea, suicidal ideation, syncope) immediately inject **STAT 112 / 911** red-flag triage guidance.
- Every structured brief returns verifiable PMID/DOI evidence pills (e.g., `PMID: 36734643`, `38114084`, `37410786`) and non-diagnostic disclaimers.

4. Message-Only Print Isolation, Binary PDF 1.4 Writer & Social Share

Implemented in `apps/web/lib/export-share.ts`, `apps/web/components/ExportPrintShareModal.tsx`, and `apps/web/app/globals.css` so users can export strictly a single response (`scope="single"`) or clean transcript messages (`scope="all"`) with zero dashboard chrome.

Pure TypeScript PDF 1.4 Engine

- `normalizeMessagesForExport()` normalizes both text-chat `Message[]` and voice `ChatRecord[]` interfaces cleanly.
- `buildClinicalPdfBlob()` constructs A4 (`595 x 842 pt`) PDF Catalog, Pages, Helvetica/Helvetica-Bold fonts, WinAnsi-safe lines wrapped at 88 chars, content streams, and exact byte-offset `xref` + `trailer`.
- Returns an authentic `application/pdf` Blob offline with zero third-party PDF bundle bloat.

Isolated Print & Outbound Sharing

- **Print Only:** Portals `#clinical-print-portal` to `document.body`, adds `.clinical-print-active`, and triggers CSS rule `body.clinical-print-active > *:not(#clinical-print-portal) { display: none !important; }`.
- **Social Intents:** `getSocialShareTargets()` builds encoded WhatsApp (`https://wa.me/?text=`), Telegram, X/Twitter, LinkedIn, Facebook, Email, and `navigator.share` File API payloads.

5. REST API Specification (`/api/v1`) & Section 10 IoT Protocol

Router Module	Key Endpoints	Contract / Behavior
Auth & Users	`POST /auth/register`, `/login`, `/refresh`, `/logout`; `GET|PATCH|DELETE /users/me`	HS256 JWT (24h access / 7d refresh rotation), Bcrypt cost-12 hash, session invalidation via `user_sessions`.
AI & Voice	`GET /assistant/providers`; `POST /assistant/conversations/{id}/messages`; `POST /voice/process`	Dispatches `buildContextualClinicalReply` returning markdown, PMIDs, `reasoning_trace`, `follow_up_questions`, and `lstm_forecast`.
Reports & Metrics	`POST /reports/upload`, `/{id}/analyze`; `GET|POST /metrics`, `/metrics/summary`, `/export/csv`	15MB MIME-validated upload, lab biomarker extraction with abnormal flags, CSV export streams.
IoT Ingestion	`POST /ingestion/sensors`, `/ingestion/wearables`, `/ingestion/iot`	Section 10 unified JSON packet schema with **SHA-256** deduplication on `device_id + measurement_type + value + timestamp_iso`.
ML & Privacy	`GET /ml/models`, `/ml/experiments`; `POST /ml/predict-risk`; `GET|PATCH /privacy/consent`; `POST /privacy/export`	Gradient Boost risk prediction with 95% CI + TreeSHAP factors; itemized DPDP consent ledger & full cascade erasure.

6. Zero-Trust Security Architecture & Threat Mitigations

Access Control, IDOR & Rate Limiting

- **IDOR Prevention:** Tenant resource endpoints (`/reports/{id}`, `/metrics/{id}`, `/records/{id}`, `/conversations/{id}`) assert `resource.user_id == current_user.id`, raising HTTP `403` on mismatch.
- **Sliding-Window Rate Limiter:** Implemented in `backend/app/main.py` across separate 60s buckets for login, assistant/voice, and report upload routes.
- **PostgreSQL RLS & ReBAC:** Forced Row-Level Security paired with relationship-based clinician access scope.

Input Safety, Privacy & Auditability

- **Prompt & Upload Defenses:** Jailbreak filter guards external dispatch; MIME allowlist restricted to `pdf`, `csv`, `txt`, png, `jpeg` (<=15MB); zero raw DNA sequencing files retained.
- **DPDP Act 2023 & GDPR Compliance:** Granular toggles (`ai_data_analysis`, `health_records_storage`, `anonymized_research`) + one-click JSON portability + cascade deletion.
- **Audit Ledger:** High-consequence events log timestamp, user ID, resource type, and client IP to `audit_events`.

7. Verification Matrix & Deployment Configuration

Verification Gate	Tooling / Command	Observed Verification Outcome
Type Safety	`npm run typecheck --workspace=apps/web`	`tsc --noEmit` exited with code `0`.
Frontend Automated Tests	`vitest run` (JSDOM + Testing Library)	**9 test files, 37/37 tests passing** (API client, auth/brand, phone OTP, genetics, ML Lab, Bi-LSTM & PDF export, landing carousel, public docs).
Production Build	`next build` (Next.js 14.2.35)	Compiled cleanly; **43/43** static pages & dynamic API route traces built.
Backend Test Suite	`pytest` (6 modules in `backend/tests/`)	Auth/security, enterprise IDOR, agents/devices, clinician/admin, medical workflows, voice service.
Production Hosting	Vercel CLI + `vercel.json` / `render.yaml` / Docker	Live alias `https://medipredict-ai-seven.vercel.app` (verified HTTP `200` on `/`, `/assistant`, `/assistant/voice`, and `/api/v1/assistant/providers`).

> **Developer Documentation Reference:** Full setup specifications remain maintained across repository guides in `medipredict-ai/docs/architecture.md`, `api.md`, `security.md`, `privacy.md`, `FULL_STACK_DEPLOYMENT_GUIDE.md`, and `hardware/README.md`.